laws of physics will be the same inside your starship as on Earth, and that when you look out at the heavens around you, you will see the entire Universe shining down at you in a brilliant, small disk of light. I guarantee that, if you send a robot probe down near the horizon of a spinning hole, blast as it may it will never be able to move forward or backward at any speed other than the hole's own spin speed (270 circuits per second in my example). I guarantee that a rapidly spinning hole can store as much as 29 percent of its mass as spin energy, and that if one is clever enough, one can extract that energy and use it.

How can I guarantee all these things with considerable confidence? After all, I have never seen a black hole. Nobody has. Astronomers have found only indirect evidence for the existence of black holes[47] and no observational evidence whatsoever for their claimed detailed properties. How can I be so audacious as to guarantee so much about them? For one simple reason. Just as the laws of physics predict the pattern of ocean tides on Earth, the time and height of each high tide and each low tide, so also the laws of physics, if we understand them correctly, predict these black-hole properties, and predict them with no equivocation. From Newton's description of the laws of physics one can deduce, by mathematical calculations, the sequence of Earth tides for the year 1999 or the year 2010; similarly, from Einstein's general relativity description of the laws, one can deduce, by mathematical calculations, everything there is to know about the properties of black holes, from the horizon on outward.

And why do I believe that Einstein's general relativity description of the fundamental laws of physics is a highly accurate one? After all, we know that Newton's description ceases to be accurate near a black hole.

Successful descriptions of the fundamental laws contain within themselves a strong indication of where they will fail.[48] Newton's description tells us itself that it will probably fail near a black hole (though we only learned in the twentieth century how to read this out of Newton's description). Similarly, Einstein's general relativity description exudes confidence in itself outside a black hole, at the hole's horizon, and inside the hole all the way down almost (but not quite) to the singularity at its center. This is one thing that gives me confidence in general relativity's predictions. Another is the fact that, although general relativity's black-hole predictions have not yet been tested directly, there have been high-precision